The Price of Exclusion: SME Set-Asides in Public Procurement

Darcio Genicolo-Martins[a]

[a] *INSPER, São Paulo*

Abstract

SME set-asides support protected firms by excluding their rivals. This paper shows why that design is costly. I study a legal reinterpretation that expanded SME-only tendering in São Paulo's electronic procurement platform and use reverse-auction drop-out bids to recover type-specific cost distributions. The set-aside changes two margins with opposite signs: it removes non-SME bidders from the price-forming pool, but it also induces SME entry and changes SME composition. A structural decomposition shows that entry responds, but exclusion dominates. Removing non-SMEs raises simulated prices by 0.371/0.565 of the reference price in non-pharmaceutical/pharmaceutical procurement; post-policy SME entry offsets -0.144/-0.256, leaving positive net effects. Welfare losses equal 28.9/44.8 percent of the open-regime price at $\lambda = 0.30$. A 10 percent SME price preference preserves non-SME participation and has near-zero price effects in thick standardized markets, but delivers less redistribution. The policy implication is a frontier, not a ban on SME support: use exclusion only when the planner is prepared to pay for the missing competitive discipline.

Keywords: public procurement, SME set-asides, auctions, structural estimation, welfare

JEL MSC: H32, H57, L26, L53, D44

Email address: `darciogm1@insper.edu.br` (Darcio Genicolo-Martins)

1. Introduction

Procurement set-asides for small and medium-size enterprises (SMEs) are meant to widen access to public contracts. They do so by changing eligibility: some contracts are reserved for small firms, and larger firms are not allowed to compete. That design creates a basic tension. A set-aside may attract protected firms into the auction, but it also removes the unprotected firms that would otherwise discipline price formation. These two forces have opposite signs. The entry response pushes prices down; the loss of competitive discipline pushes prices up. The welfare case for a set-aside depends on which force dominates.

This paper estimates that decomposition. The question is not whether SME set-asides raise procurement prices; a large empirical literature already shows that bidder-preference and set-aside programs can be costly (Marion, 2007; Krasnokutskaya, 2011; Athey et al., 2013; Nakabayashi, 2013; Hyytinen et al., 2018). The question is why they are costly, and what that implies for policy design. Reduced-form price effects cannot answer that question because a set-aside changes two margins at once: it excludes non-SMEs from the price-forming bidder pool and it changes SME participation and composition. A policy that supports SMEs while keeping non-SMEs inside the auction, such as a price preference, has a different economic logic from a policy that supports SMEs by removing their rivals. The relevant design question is therefore not simply whether the government should support SMEs. It is whether SME support should preserve or remove the bidders that discipline the auction.

The empirical setting is São Paulo's centralized electronic procurement platform, the Bolsa Eletrônica de Compras (BEC). In late 2017 and early 2018, a legal reinterpretation shifted how the state applied Brazil's SME-only procurement rule. The old interpretation applied the eligibility threshold to the total value of a purchase notice; the new interpretation applied it item-by-item. That change mechanically exposed medical and hospital supplies (Group 65) to SME-only tendering because these purchases often combine many small items inside large purchase notices. The empirical cutoff is March 2018, when Group-65 public buyers began mass take-up of the new functionality. The trigger was legal and administrative rather than a medical-supply-specific policy initiative, and it changed admissibility rather than the production technology of the goods.

The first-stage facts line up with the institutional story. After the cutoff, SME-only adoption in Group 65 rises sharply, non-SME participation falls, and SME participation roughly doubles. Winning prices also rise. A difference-in-differences benchmark against 76 never-treated product groups estimates that absolute winning prices increase by about 10–11 percent in the structural window. I use that reduced-form evidence as a sign, timing, and scale benchmark. The paper's main object is structural: separating the price increase into the loss of non-SME competitive discipline and the offsetting entry/composition response of SMEs.

The auction format makes that decomposition feasible. BEC's main procurement format is the electronic *Pregão*, an English-reverse auction in which firms submit successively lower offers until only one remains. Under independent private values, losing bidders' drop-out prices reveal their costs (Haile and Tamer, 2003). This feature is valuable in a setting where the policy changes the composition of the bidder pool. It allows the paper to recover type-specific cost distributions for SMEs and non-SMEs, correct for auction-level scale heterogeneity in the spirit of Krasnokutskaya (2011), and simulate counterfactual auctions under observed equilibrium entry. The exercise is deliberately weaker than a full entry equilibrium: observed pre- and post-policy entry rates are taken as primitives, and the post-policy SME cost distribution is estimated from post-policy exits rather than derived from a formal selection model. This is an accounting decomposition under explicit auction assumptions, not a complete model of firm participation.

Three counterfactual prices summarize the paper. Let S_1 be the open auction with the pre-policy bidder pool, S_2 the SME-only auction holding the pre-policy SME pool fixed, and S_3 the SME-only auction with the observed post-policy SME pool. Then

$$p_{S_3} - p_{S_1} = \underbrace{p_{S_2} - p_{S_1}}_{\text{lost competitive discipline}} + \underbrace{p_{S_3} - p_{S_2}}_{\text{entry/composition offset}} .$$

The first term is the cost of removing non-SMEs from the price-forming pool. The second term is the offset created by the post-policy SME pool. This decomposition is the paper's organizing object. It turns a set-aside price effect into a design question: how much of the cost comes from replacing competition by eligibility, and how much is recovered by protected entry?

The main result is that entry responds, but exclusion dominates. In the baseline structural simulation, moving from the open auction to an SME-only auction at the pre-policy SME pool raises the simulated clearing price by 0.371 of the reference price in non-pharmaceutical supplies and 0.565 in pharmaceuticals. Allowing the post-policy SME pool to enter offsets part of that shock, by -0.144 and -0.256 respectively. The net effect remains positive: 0.227 in non-pharmaceuticals and 0.309 in pharmaceuticals. Relative to the net effect, the exclusion component is super-additive, 164 percent in non-pharmaceuticals and 183 percent in pharmaceuticals, because SME entry partially undoes the initial shock. In absolute-magnitude terms, the exclusion component accounts for 72.0 and 68.8 percent of the two components. The interpretation is simple: the policy attracts SMEs, but it does not recreate the competitive discipline supplied by excluded non-SMEs.

This mechanism changes the policy comparison. A full set-aside supports SMEs by removing their rivals. A price preference supports SMEs while leaving non-SMEs inside the auction. The difference matters because non-SMEs can remain price-forming even when the scoring rule favors SMEs. I compare the observed SME-only design to a 10 percent SME price preference using the same recovered cost distributions and entry primitives. In thick, standardized non-pharmaceutical markets, the price preference has near-zero price and welfare cost while the full set-aside generates a substantial welfare loss. The preference is not a free version of the set-aside: it delivers a smaller redistributive intervention. The correct comparison is a frontier. The set-aside is a high-cost, high-redistribution instrument; the price preference is a low-cost, lower-redistribution instrument that preserves competitive discipline. A social planner should prefer the set-aside only if the additional SME surplus is worth the allocative and fiscal cost of removing non-SMEs from the auction.

The strongest policy result is in thick standardized markets. In thinner and more heterogeneous pharmaceutical markets, the ranking depends on how the post-policy SME pool is modeled. The main specification estimates the post-policy SME cost distribution directly from post-policy exits; a strict invariance benchmark instead imposes that the SME cost distribution is the same before and after the policy. The non-pharmaceutical ranking is stable across these readings. The pharmaceutical ranking is not. I treat that difference as a scope condition rather than as a second headline: bidder exclusion is hardest to justify where the protected pool is thick enough for a preference rule to tilt allocation without removing competitive discipline, and most defensible, if at all, where the protected pool is thin, the good is heterogeneous, and the planner places high value on SME surplus.

The paper makes three contributions. First, it decomposes the price effect of SME set-asides into two opposing channels: the loss of competitive discipline from excluding non-SMEs and the protected-entry/composition offset. Second, it uses the cost-revealing structure of electronic reverse auctions to estimate that decomposition from drop-out bids in a real procurement reform, linking a legal shock to structural price formation without relying on first-price bid-function inversion. Third, it recasts the set-aside-versus-preference comparison as a question of preserving competition inside the auction. This connects the empirical procurement-preference literature (Marion, 2007; Krasnokutskaya, 2011; Krasnokutskaya and Seim, 2011; Athey et al., 2013) to a public-finance welfare-weight interpretation in which transfers to SMEs, allocative waste, and the marginal cost of public funds are kept separate (Saez and Stantcheva, 2016).

The broader lesson is that eligibility is a costly way to buy redistribution when excluded firms are the ones disciplining the auction. Set-asides can be defensible, but their defense must be explicit: the planner must value the additional SME surplus enough to justify the allocative and fiscal cost of removing rival bidders. Where protected firms are already numerous and goods are standardized, that case is hard to make. The natural benchmark is therefore not no SME support, but SME support that keeps competition in the room.

2. Setting, Data, and First-Stage Facts

This section establishes the institutional shock and the first-stage facts that the decomposition uses. The reform changed which firms could compete for eligible Group 65 items. In the data, the change raises SME-only take-up, reduces non-SME participation, increases SME participation, and is associated with higher winning prices. Those facts are not the paper's mechanism; they are the reduced-form footprint of the mechanism estimated in the structural sections.

2.1. Legal shock

Brazil's federal SME statute, LC 123/2006, requires exclusive SME tenders for procurement items valued at R$ 80,000 or less. The 2014 amendment, LC 147/2014, made the SME-only rule mandatory rather than optional. For most goods on São Paulo's BEC platform, that rule already operated before the period studied here. Group 65—medical, dental, and hospital equipment and supplies—was different for a legal reason. The prevailing interpretation applied the R$ 80,000 threshold to the total value of the purchase notice, not to each item or lot. Because medical-supply notices often bundle many small items inside large purchase orders, the total-notice interpretation mechanically kept Group 65 outside the SME-only default even when individual items were below the threshold.

The relevant change was a generic legal reinterpretation. BEC enabled SME-only purchase-order functionality in COMUNICADO BEC nº 02/2017 on 18 July 2017 and mixed-exclusivity notices in COMUNICADO BEC nº 03/2017 on 20 November 2017. PGE-SP then issued Parecer Sub-G Cons. nº 151/2017 on 11 December 2017, holding that the R$ 80,000 threshold applies item-by-item regardless of the total value of the purchase offer. The empirical cutoff is March 2018, when Group 65 public buyer units began mass take-up of the functionality. TCE-SP aligned with the new interpretation on 16 May 2018, after the empirical cutoff, ratifying rather than initiating the operational sequence.

This timing matters for identification. The trigger was not a Group 65 market intervention and did not speak to medical-supply prices, supplier complaints, or technology. It was a general reading of procurement law that interacted with the way Group 65 notices were written. The institutional claim is therefore narrow: the cutoff shifts admissibility and the composition of the active bidder pool. It does not by itself identify a full entry model, and it does not preclude composition changes among SMEs. Those are treated explicitly in the structural decomposition and robustness checks.

2.2. Data and auction format

The data are administrative microdata from BEC, São Paulo's centralized electronic procurement platform. The raw extract contains 3.7 million bid-level observations across 82,569 items, 832,984 purchase orders, and 1,344 public buyer units. Group 65 accounts for roughly 27 percent of platform transactions in the window used here and is the treated group. The reduced-form benchmark compares Group 65 to 76 never-treated product groups. The structural exercise uses Group 65 Pregão auctions in a symmetric 18-month window on each side of the March 2018 cutoff, from September 2016 through August 2019.

The Pregão format is central. It is an electronic English-reverse auction: the auction opens at a buyer reference price, firms submit lower offers, and firms exit as the clock falls. Under the independent-private-values interpretation used in the model, losing drop-out prices are cost observations. For that reason, the structural sample restricts attention to Pregão auctions with at least 2 active firms and normalized bids $c_\epsilon = b/p^{\text{ref}} \in (0, 3]$. The resulting sample contains 297,967 firm-auction observations across 97,993 auctions, split almost evenly between the pre-period (48,740 auctions) and post-period (49,253 auctions).

Each observation is classified by bidder type and product class. SME status is taken from the BEC administrative registry and validated against historical firm bidding records. I split Group 65 into non-pharmaceutical supplies and a pharmaceutical class covering CMED-regulated medicines and biologicals. This split is not a second treatment. It is a market-thickness distinction: the non-pharmaceutical segment is closer to a thick standardized market, while pharmaceutical procurement has fewer SME bidders and more within-class heterogeneity. The main policy claim is strongest in the former; the latter is used as a boundary case for external validity.

2.3. First-stage facts

Table 1 reports the participation facts that motivate the structural decomposition. In Pregão non-pharmaceutical auctions, the average number of SME bidders rises from 0.94 before the cutoff to 1.87 after it, while non-SME participation falls from 2.68 to 1.50. In pharmaceutical auctions, SME participation rises from 0.55 to 1.22, while non-SME participation falls from 2.61 to 1.66. The post-period non-SME counts are not zero because Group 65 take-up is incomplete: state-level SME-only adherence in Group 65 reaches 43 percent after the cutoff rather than full coverage. The policy nevertheless moves both margins in the direction required by the paper's mechanism: non-SMEs become less present in the treated market and SMEs become more present.

Table 1: First-stage participation facts in Group 65 Pregão auctions

	Non-pharma	Pharma
Pre: SME bidders per auction	0.94	0.55
Pre: non-SME bidders per auction	2.68	2.61
Post: SME bidders per auction	1.87	1.22
Post: non-SME bidders per auction	1.50	1.66
Change in SME bidders	+0.93	+0.67
Change in non-SME bidders	-1.18	-0.95

Sample: Group 65 Pregão auctions in the 18-month window on each side of the March 2018 cutoff, after the structural filters $c_\epsilon \in (0, 3]$ and at least 2 active firms. Counts are average distinct bidders per auction by type and class. Post-period non-SME counts remain positive because the empirical SME-only take-up rate is 43 percent, not 100 percent.

Prices move in the same period. Table 2 reports a compact difference-in-differences benchmark. The estimating equation compares Group 65 to never-treated product groups, with item and month fixed effects, auction-format and quantity controls, and item-clustered standard errors. The coefficient is written in the paper's "DiD-in-reverse" convention: a negative coefficient on Group 65 interacted with the pre-period means that the open regime was cheaper, so the post-cutoff SME-only extension undoes that discount. Across windows, the log-price coefficient is between -0.108 and -0.148 with PBU controls; the 18-month benchmark is -0.113. Exponentiating the magnitude gives the 10–11 percent price benchmark used in the introduction.

Two qualifications are important. First, the reduced-form design is not asked to do the paper's full work. Group 65 differs from the pooled controls in several pre-period characteristics, and modern DiD estimators produce less precise estimates than the two-way fixed-effect benchmark. I therefore use the DiD as evidence on sign, timing, and approximate scale rather than as the final estimand. Second, the first-stage facts already

Table 2: Reduced-form price benchmark

Window around cutoff	6 months	12 months	18 months
$g65 \times Pre$ on $\log p^{\text{final}}$	-0.148	-0.108	-0.113
Standard error	(0.014)	(0.013)	(0.012)
Observations	219,535	439,054	649,714

Estimates use item and month fixed effects, controls for sealed-bid format and log quantity, PBU controls, and item-clustered standard errors. The coefficient is reported in the paper's sign convention: a negative pre-period coefficient is the open-regime price discount that disappears after the SME-only extension. The DiD is used as a sign, timing, and scale benchmark; the structural sections estimate the mechanism.

imply that a price regression cannot identify the mechanism. The same legal shock both removes non-SMEs from the eligible pool and induces additional SME participation. The rest of the paper uses the Pregão cost information to separate those margins.

3. Model and Identification

The empirical facts in Section 2 show that the reform moves participation and prices in the direction implied by the institutional shock. This section defines the structural objects used to separate the two channels. The model is intentionally spare. It uses the auction format to recover type-specific costs, takes observed entry as an equilibrium object, and simulates three counterfactual bidder pools. The goal is not to estimate a full entry equilibrium; it is to decompose the price effect of exclusion into lost competitive discipline and the offset created by protected entry and composition.

3.1. Auction environment

Consider a single procurement auction for one indivisible contract. Each firm i has type $k_i \in \{\text{SME}, \neg\}$ and draws a private supply cost c_i from a type-specific distribution F_c^k. The two distributions are allowed to differ. The buyer runs an electronic Pregão, an English-reverse auction in which the price clock moves downward from the buyer's reference price and firms remain active until they are no longer willing to supply at the current price. The last active firm wins and supplies the contract at the clearing price.

Under the independent-private-values interpretation, exit at cost is weakly dominant in this format (Haile and Tamer, 2003). A losing firm's drop-out price is therefore a direct observation of its cost. This is the key identifying advantage of the setting. In first-price procurement auctions, the researcher observes strategically shaded bids and must recover costs by inverting a bid function. Here the price-forming statistic can be recovered from exits. The winner is censored because the winner does not drop out; its cost is known only to be no higher than the second-lowest exit. The baseline estimator includes the winner's final price as a tight upper-bound observation, while the robustness section reports a Turnbull NPMLE that treats the winner as left-censored at that bound.

The expected transaction price in the clock interpretation is the expected second-lowest cost, $E[c_{(2)}]$, while the production cost of the allocation is the winner's cost, $E[c_{(1)}]$. This distinction matters later for welfare: $c_{(2)}$ determines the government's payment, while $c_{(1)}$ determines allocative efficiency.

3.2. Recovering cost primitives

The structural sample is organized by class, period, and bidder type. For each cell, loser drop-outs identify the corresponding empirical distribution of normalized costs. Costs are normalized by the buyer's reference price, so the object entering the simulation is a distribution of c_i/p^{ref}. Reference-price normalization absorbs the main cross-item scale differences, but it does not remove all auction-level heterogeneity. Two auctions in the same product class can differ in volume, delivery burden, regulatory documentation, or item-specific complexity in ways that move every bid in the auction together.

I therefore use the standard multiplicative auction-level heterogeneity correction of Krasnokutskaya (2011). Let

$$y_{it} = \log(b_{it}/p_t^{\text{ref}}) = a_t + e_{it},$$

where a_t is an auction-level scale component and e_{it} is the bidder-specific cost component. A method-of-moments variance decomposition estimates the between-auction and within-auction components, and a best-linear-predictor shrinkage removes $\hat{a}_t$ from each observation. The simulation uses the cleaned residual cost distribution $\exp(\hat{e}_{it})$. In the v6 estimates, the resulting Pregão intraclass correlations range from 0.36 to 0.59 in the main cells, with values comparable to procurement settings where this correction is standard.

Two diagnostics discipline the cost-recovery step. First, in a pre-policy cell unaffected by the set-aside, the cost distribution recovered from Pregão drop-outs can be compared to a GPV recovery from Convite first-price bids (Guerre et al., 2000). The two recoveries are close in the pharma non-SME pre-period cell, which is the cleanest falsification cell because those firms are not targeted by the policy. Second, the Turnbull NPMLE for winner censoring delivers the same decomposition qualitatively: the within-auction share remains large (74.0 percent in non-pharma and 82.0 percent in pharma), and the total effect differs from the baseline by at most 16 percent across the reported cells. These checks do not prove IPV. They show that the main cost-recovery step is not being driven by winner censoring or by an obvious conflict between auction formats.

3.3. Entry and the post-policy SME distribution

Entry is treated as observed equilibrium entry. For each class and period, I use the observed average numbers of SME and non-SME bidders per auction as the arrival rates for the simulation. In the open pre-policy regime, both SMEs and non-SMEs may enter. In the SME-only counterfactual, non-SMEs are removed from the eligible pool. The observed post-policy SME count is used for the post-policy SME-only scenario.

This choice is deliberate. A full entry model would require assumptions on fixed costs, information, bidder-specific participation decisions, and how the legal shock changes expected profits. The legal shock directly identifies admissibility and observed participation, not a free-entry primitive. The paper therefore follows the reduced-form-entry spirit of Athey et al. (2011): observed pre- and post-policy bidder counts are equilibrium objects used in counterfactual price formation.

The same caution applies to the post-policy SME cost distribution. The baseline simulation estimates $F_c^{\text{SME,Post}}$ directly from post-policy drop-outs and uses it as the cost distribution of active post-policy SME bidders. This object may reflect selection into the protected market, technology or sourcing changes within the SME segment, or other composition changes around the cutoff. The baseline does not derive that difference from a formal selection model. The strict-invariance benchmark instead imposes $F_c^{\text{SME,Post}} = F_c^{\text{SME,Pre}}$. The two readings bracket how much the policy conclusion depends on the empirical post-policy SME distribution. The non-pharmaceutical policy ranking is stable across them; the pharmaceutical ranking is not, which is why pharma is treated as a boundary case rather than as the headline.

3.4. Counterfactual objects

The decomposition uses three simulated auction environments:

- S_1: open auction, pre-policy bidder pool. SMEs and non-SMEs enter at their observed pre-policy rates, with pre-policy cost distributions.
- S_2: SME-only auction, pre-policy SME pool. Non-SMEs are removed, while the SME arrival rate and SME cost distribution are held at their pre-policy values.
- S_3: SME-only auction, post-policy SME pool. Non-SMEs are removed, and SMEs enter with the observed post-policy arrival rate and the baseline post-policy SME cost distribution.

For each scenario, I draw bidder counts from the corresponding type-specific arrival process, draw costs from the recovered cost distributions, sort the draws, and record $c_{(1)}$

and $c_{(2)}$. The simulated price is $p_S = E_S[c_{(2)}]$, normalized by the reference price. The simulated production cost is $E_S[c_{(1)}]$.

The total set-aside price effect is

$$\Delta^{SA} = p_{S_3} - p_{S_1}.$$

It decomposes mechanically into two terms:

$$p_{S_3} - p_{S_1} = \underbrace{p_{S_2} - p_{S_1}}_{\Delta^{excl} \text{ lost competitive discipline}} + \underbrace{p_{S_3} - p_{S_2}}_{\Delta^{entry} \text{ entry/composition offset}}. \tag{1}$$

The first term holds the SME pool fixed and removes non-SMEs. It captures the price-forming cost of bidder exclusion: the second-order statistic is now drawn from the protected pool rather than from the full pool. The second term allows the post-policy SME pool to enter. It captures the extent to which protected entry and composition offset the exclusion shock. This term can be negative, and in the data it is: additional SME participation partially restores competition, but not enough to undo the price increase.

The labels in equation (1) are intentionally descriptive rather than behavioral. In an English-reverse IPV auction, surviving bidders do not need to shade bids differently when rivals are excluded; exit at cost remains weakly dominant. The lost-discipline component is therefore an order-statistic effect of admissibility, not evidence that SMEs bid less aggressively. This is the distinction between the mechanism in this paper and the first-price settings in which bidder preferences combine cost differences, order statistics, and strategic bid shading.

3.5. Identification threats and what disciplines them

Four threats matter for the interpretation. The first is the IPV/drop-out restriction. If drop-outs are not informative about costs, the recovered F_c^k distributions are not structural cost primitives. The cross-modality check and the Turnbull censoring exercise are the main diagnostics, and Section 6 summarizes an affiliation sensitivity that allows within-auction cost correlation up to $\rho_c = 0.3$.

The second threat is primitive stability. The institutional shock shifts admissibility, but the recovered SME distribution changes across periods. The baseline treats that empirical post-policy distribution as the relevant active SME pool; strict invariance imposes no change. The comparison between those readings is central to the scope conditions.

The third threat is entry aggregation. The baseline uses average class-by-period arrival rates. That is enough for the main accounting exercise, but it is coarse. Section 6 therefore treats empirical entry-count distributions and finer arrival specifications as robustness checks rather than as new mechanisms.

The fourth threat is coordination. Bidder-pair screens detect clustering in the data, so the interpretation of drop-out prices as costs requires care. The important diagnostic is whether clustering intensifies after non-SMEs are removed, and whether the decomposition survives in low-coordination samples. The paper treats this as a first-order threat to interpretation, not as a cosmetic robustness exercise.

4. Exclusion, Entry, and Price Formation

The set-aside changes two margins at once. It removes non-SMEs from eligible auctions, but it also attracts more SMEs into the protected pool. This section uses the counterfactuals from Section 3 to separate the two forces. The result is simple: SME entry responds strongly, but the price effect is dominated by the loss of non-SME competitive discipline.

4.1. Entry responds

The first-stage participation facts already rule out a simple story in which prices rise because protected firms fail to enter. In non-pharmaceutical Pregão auctions, the average number of SME bidders rises from 0.94 to 1.87 per auction after the cutoff. In pharmaceutical auctions, it rises from 0.55 to 1.22. These are large changes relative to the pre-policy protected pools. Non-SME participation moves in the opposite direction, falling from 2.68 to 1.50 in non-pharma and from 2.61 to 1.66 in pharma.

That participation response matters for interpretation. If one held the SME pool fixed at its pre-policy size, the set-aside would look more expensive than the realized post-policy environment. The government gets some competition back through induced SME participation. The key question is whether that offset is large enough to replace the price discipline supplied by the excluded non-SMEs.

4.2. Exclusion dominates

Table 3 reports the three simulated price statistics. The open pre-policy auction is S_1. The SME-only auction holding the pre-policy SME pool fixed is S_2. The SME-only auction with the observed post-policy SME pool is S_3. Prices are normalized by the buyer's reference price and correspond to the simulated second-order statistic, $E[c_{(2)}]$.

The first move, $S_1 \to S_2$, isolates bidder exclusion. In non-pharmaceutical markets, removing non-SMEs while holding the pre-policy SME pool fixed raises the simulated price from 0.774 to 1.144, an increase of 0.371 of the reference price. In pharmaceutical markets, the same move raises the simulated price from 0.654 to 1.219, an increase of 0.565. This is the price-forming cost of replacing the full bidder pool with the protected pool.

The second move, $S_2 \to S_3$, allows the post-policy SME pool to enter. This term is negative in both classes. In non-pharma, SME entry/composition reduces the simulated price by -0.144 of the reference price. In pharma, it reduces the simulated price by -0.256. The realized set-aside effect, $S_3 - S_1$, remains positive: 0.227 in non-pharma and 0.309 in pharma. Entry attenuates the price shock; it does not eliminate it.

The relative-to-net column is intentionally reported because it shows why the ab-

Table 3: Price decomposition: exclusion versus entry/composition

	Simulated price			Decomposition			
Class	S_1	S_2	S_3	$S_2 - S_1$	$S_3 - S_2$	$S_3 - S_1$	Excl./net
Non-pharma	0.774	1.144	1.000	+0.371	-0.144	+0.227	164%
Pharma	0.654	1.219	0.963	+0.565	-0.256	+0.309	183%

S_1 is the open auction with the pre-policy bidder pool. S_2 is the SME-only auction holding the pre-policy SME pool fixed. S_3 is the SME-only auction using the observed post-policy SME pool. Prices are simulated $E[c_{(2)}]$ values normalized by the buyer reference price. "Excl./net" reports $(S_2 - S_1)/(S_3 - S_1)$. The entry/composition term is negative because the post-policy protected pool partially offsets the exclusion shock.

solute share can understate the size of the exclusion margin. In non-pharma, the exclusion component is 164 percent of the net effect; in pharma it is 183 percent. The entry/composition term offsets -64 percent and -83 percent of the net effect, respectively. If instead one normalizes by the sum of absolute component magnitudes, exclusion accounts for 72.0 percent in non-pharma and 68.8 percent in pharma. Both normalizations point in the same direction: the dominant force is the loss of the non-SME price-forming pool, not a lack of SME entry.

Figure 1 makes the same point visually. The jump from S_1 to S_2 is the cost of exclusion. The decline from S_2 to S_3 is the offset from the realized protected pool. The remaining gap between S_1 and S_3 is the realized price cost of the set-aside under observed entry.

4.3. Entry and composition are offsets, not the source

The decomposition separates two facts that are often conflated. The set-aside does induce protected entry. That entry is valuable in price terms because it pulls the simulated price down from S_2 to S_3. But the policy first creates a larger price increase by removing non-SMEs from the auction. In the baseline simulation, holding the protected pool at its pre-policy size would make the latent price shock 59 percent larger in non-pharma and 81 percent larger in pharma. The realized post-policy SME pool absorbs part of that shock, but the net effect remains positive in both classes.

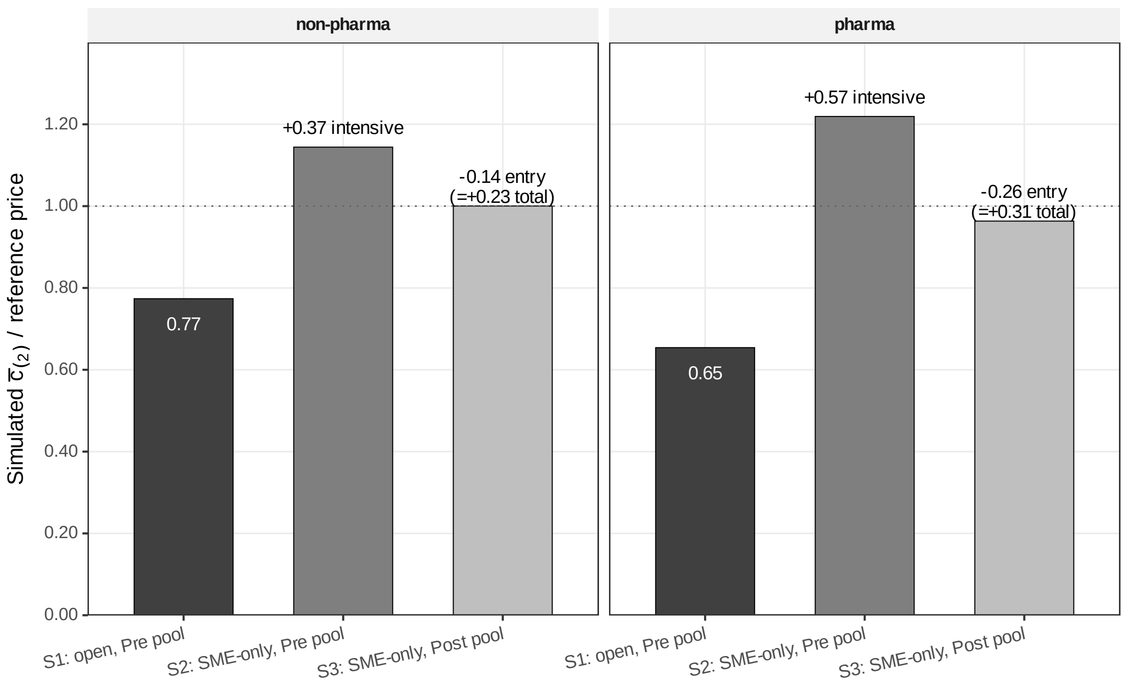

Figure 1: *Counterfactual price decomposition.* Bars report the simulated second-order statistic normalized by the buyer reference price. S_1 is the open pre-policy auction, S_2 removes non-SMEs while holding the pre-policy SME pool fixed, and S_3 allows the observed post-policy SME pool to enter. The movement from S_1 to S_2 is the lost competitive-discipline component; the movement from S_2 to S_3 is the entry/composition offset.

The entry/composition term should not be overinterpreted as pure entry. It also contains whatever empirical difference exists between the active SME bidders observed before and after the cutoff. That is why the paper treats $F_c^{\text{SME,Post}}$ explicitly as an observed post-policy active-bidder distribution rather than as the output of a formal selection model. Two diagnostics keep the conclusion focused. First, the decomposition remains exclusion-dominant under the Turnbull cost estimator, with exclusion shares of 74.0 percent in non-pharma and 82.0 percent in pharma on the absolute-share normalization. Second, under the strict-invariance benchmark that forces the post-policy SME distribution to equal the pre-policy one, the exclusion component remains the larger component (85 percent in non-pharma and 79 percent in pharma). What changes under strict invariance is the policy ranking in pharma, not the basic price-formation fact that bidder exclusion is the larger force.

Firm turnover helps explain why the pharma ranking is more fragile. In pharmaceuticals, 61.9 percent of post-policy SME firms are new relative to the pre-period pool and they account for 37.8 percent of post-policy SME bids. In non-pharma, the corresponding bid share is 23.0 percent. The composition shift is therefore larger precisely in the class where the welfare ranking is model-sensitive. This is a scope condition, not a competing headline. The robust result is that in the thicker non-pharmaceutical market, the price cost is driven by removing non-SMEs and survives alternative treatments of the protected pool.

4.4. Interpretation

The mechanism is not that protected firms strategically bid less aggressively after the set-aside. In the maintained Pregão/IPV model, bidders exit at cost. The mechanism is that the set-aside changes which cost order statistic forms the price. In the open auction, non-SMEs remain in the pool and discipline the second-lowest cost. In the SME-only auction, that discipline is removed. Additional SME entry pushes back against the loss, but the post-policy protected pool does not recreate the full-pool benchmark.

This is the price-formation result the welfare section builds on. A policy that supports SMEs by excluding non-SMEs buys redistribution by weakening the auction's price-forming pool. A policy that supports SMEs while leaving non-SMEs in the auction works through a different channel: it can tilt allocation toward SMEs while preserving the bidders that discipline the price.

5. Welfare and Policy Design

The price decomposition has a direct welfare interpretation. A set-aside raises government outlays because it changes the order statistic that determines payment, but the social cost is not the price effect itself. Part of the extra payment is a transfer to the protected winner; part reflects a real production inefficiency from assigning the contract to a higher-cost firm; and part is the tax distortion created by financing the additional public outlay. The policy question is therefore sharper than whether SME support is desirable. It is whether the incremental SME surplus generated by excluding rival bidders is valuable enough to justify the allocative and fiscal cost of removing them from the auction.

5.1. Welfare accounting

Let p^V denote the expected procurement price under policy V, and let $c_{(1)}^V$ denote the winner's production cost. Relative to the open benchmark S_1, the per-auction welfare loss from the SME-only set-aside V_0 is

$$\mathrm{Loss}(V_0) = \underbrace{c_{(1)}^{V_0} - c_{(1)}^{S_1}}_{\mathrm{DWL_{alloc}}} + \underbrace{\lambda\left(p^{V_0} - p^{S_1}\right)}_{\text{MCPF distortion}}. \tag{2}$$

The first term is allocative deadweight loss: the additional production cost created when the restricted auction selects an SME whose cost exceeds the lowest-cost firm in the full pool. The second term prices the marginal cost of public funds; the baseline uses $\lambda = 0.30$, consistent with the standard public-finance range used to evaluate tax-financed spending (Ballard et al., 1985; Hendren and Sprung-Keyser, 2020; Finkelstein and Hendren, 2020). The remaining component of the government's extra payment is not destruction in this accounting. It is an implicit transfer to SME bidders, and it becomes a social benefit only to the extent that the planner places extra welfare weight on SME surplus.

5.2. The welfare cost of exclusion

Table 4 summarizes the welfare accounting and the non-exclusionary policy benchmark. In non-pharmaceuticals, the full set-aside raises government payments by 0.247 of the reference price. Of this, 0.148 is real allocative waste and 0.074 is the MCPF distortion at $\lambda = 0.30$, yielding a welfare loss equal to 28.9 percent of the open-regime price. In pharmaceuticals, the same calculation is larger: the payment increase is 0.298, the allocative loss is 0.207, the MCPF distortion is 0.089, and the total welfare loss is 44.8 percent of the open-regime price. The lower endpoints of the bootstrap intervals remain economically large (20.5 percent in non-pharmaceuticals and 34.9 percent in pharmaceuticals), so the welfare conclusion is not a small-sample artifact.

Table 4: Welfare cost of exclusion and the price-preference benchmark

	Full SME set-aside V_0				10% preference V_3	
Class	Δ_{gov}	$\text{DWL}_{\text{alloc}}$	MCPF	Loss / p_{S_1}	Δp	SME win gain
Non-pharma	0.247	0.148	0.074	28.9%	-0.004	4.3 pp
Pharma	0.298	0.207	0.089	44.8%	+0.002	1.4 pp

V_0 is the full SME-only set-aside under the observed post-policy SME pool. Δ_{gov}, $\text{DWL}_{\text{alloc}}$, MCPF, and Δp are reported in reference-price units. MCPF uses $\lambda = 0.30$. V_3 is an open auction with a 10 percent SME price preference for winner selection; all firms remain eligible, and the government pays the actual winning bid. SME win gain is relative to the open benchmark S_1.

The magnitudes are large enough to matter outside the simulated auction. At the observed 43 percent Group-65 adherence rate, the per-auction losses imply an annual welfare cost of roughly R$ 55 million in São Paulo Group-65 procurement alone; across the realistic adherence range of 30–70percent, the range is R$ 38–89 million. These figures are not the central identification result, but they discipline the policy relevance of the mechanism: exclusion is not a symbolic preference. It is a costly way to transfer surplus.

5.3. The policy frontier

The clean benchmark is a price preference rather than no SME support. Under V_3, SMEs receive a 10 percent scoring advantage for winner selection, but non-SMEs continue to participate and continue to discipline the price-forming order statistic. The simulated price effect is essentially zero: -0.004 of the reference price in non-pharmaceuticals and +0.002 in pharmaceuticals. The instrument does not replicate the full redistribution of V_0; the SME win-rate rises by 4.3/1.4 percentage points rather than being forced to 100 percent. That difference is exactly the trade-off the set-aside hides. Price preferences buy some reallocation toward SMEs while preserving the competitors that set the auction's competitive outside option; set-asides buy more reallocation by removing that outside option.

This frontier is particularly informative because the low-cost region is not a knife-edge at 10 percent. In the preference grid, welfare costs remain inside Monte Carlo noise up to 15 percent in non-pharmaceuticals and 25 percent in pharmaceuticals, and even a 30 percent preference costs only 4.99/1.92 percent of p_{S_1}. The set-aside is therefore not the natural default instrument in thick standardized procurement. It is an extreme point on the policy frontier: maximum formal eligibility protection, purchased by deleting rival bidders from the auction.

5.4. How much must the planner value SME surplus?

The distributional case for exclusion can be stated as a welfare-weight condition. Following the social marginal welfare weight logic in Saez and Stantcheva (2016), let

w^{SME} be the planner's weight on one real of SME producer surplus relative to one real of general surplus. The planner is indifferent between the set-aside and the 10 percent preference when

$$w_{\star}^{\mathrm{SME}} = \frac{\mathrm{Loss}(V_0) - \mathrm{Loss}(V_3)}{\mathrm{SMESurplus}(V_0) - \mathrm{SMESurplus}(V_3)}. \tag{3}$$

Because V_3 has near-zero welfare cost, the numerator is essentially the deadweight cost of exclusion. The denominator is the additional SME surplus created by moving from a price preference to a full set-aside. Under the main specification, the implied indifference weights are 2.42 in non-pharmaceuticals and 2.61 in pharmaceuticals. In words, the planner must value an extra real of SME surplus at more than twice a real of general surplus to prefer excluding rival bidders over preserving them with a price preference.

The strict-invariance benchmark clarifies the scope of that statement. In non-pharmaceuticals, the preference rule continues to dominate the set-aside; the implied weight remains above the utilitarian benchmark. In pharmaceuticals, the ranking is sensitive to whether the post-policy SME pool is interpreted as a selected equilibrium participant pool or forced to have the same primitive cost distribution as the pre-policy pool; under strict invariance the implied weight falls to 0.7. That sensitivity is not a nuisance detail. It is the policy boundary of the paper: the argument against bidder exclusion is strongest where the protected pool is thick enough that redistribution can be achieved without changing the price-forming bidders.

5.5. When is exclusion defensible?

The results do not imply that set-asides are never defensible. They imply that the defense must be explicit. Exclusion becomes more plausible when the protected pool is thin, when the planner's objective is to create protected market access rather than merely tilt marginal awards, when dynamic SME capacity gains are expected to be large, or when the planner places a high social weight on SME surplus. Those conditions are not identified by the legal threshold alone. They are market-structure conditions.

Non-pharmaceutical Group-65 procurement is the hard case for the set-aside: the SME pool is thick, goods are standardized, and a price preference preserves competition while delivering SME wins at almost no welfare cost. Pharmaceuticals are the boundary case: the SME pool is thinner, composition changes more under the policy, and the welfare ranking depends on how the post-policy SME distribution is interpreted. The broader design principle is therefore conditional rather than universal. Use exclusion only when the planner is prepared to pay for the missing competitors; otherwise, support SMEs with an instrument that keeps those competitors in the auction.

6. Robustness to Core Threats

The robustness exercise is organized around the mechanism, not around a table inventory. The paper's empirical chain has four possible weak points: the legal shock may not isolate the relevant timing; drop-out bids may not recover the cost primitives needed for the order-statistic comparison; the post-policy SME pool may mix entry and selection in a way the model does not structurally derive; and the restricted auctions may intensify coordination among the remaining bidders. The checks below ask whether any of these threats overturns the central result that bidder exclusion, rather than SME entry, is the main source of the price increase.

6.1. Reduced-form timing and scale

The reduced-form evidence is useful for timing and scale, but it is not asked to carry the mechanism. The 18-month price benchmark in Table 2 is -0.113 in the paper's DiD-in-reverse convention, implying a 10–11 percent price movement. The estimate is stable across the 6-, 12-, and 18-month windows, and the legal timing survives a direct adoption-window check: excluding the BEC enablement months before the mass-take-up cutoff changes the coefficient from -0.109 to -0.087 with a standard error of 0.012.

Placebo cutoffs before the reform are smaller in magnitude (-0.013, -0.030, and -0.034) than the real-cutoff estimates.

The conservative reading is still necessary. Group 65 is not perfectly balanced against the pooled controls: 7 of 9 pre-period covariates exceed the 0.10 normalized-difference attention threshold, and 4 exceed 0.25. Modern DiD variants also reduce precision: the BJS imputation estimate is -0.0560 and the Callaway–Sant'Anna estimate is -0.0165, both weaker than the two-way fixed-effect benchmark. This is why the reduced form is deliberately demoted in the paper. It validates that prices move at the policy cutoff and that the movement has the right order of magnitude; it does not identify the exclusion and entry-composition channels.

6.2. Cost recovery and auction primitives

The structural result would be fragile if it depended on a particular treatment of winner censoring or on an implausible cost-recovery assumption. It does not. The main estimator uses all observed Pregão drop-out information, while a Turnbull NPMLE treats the winner as left-censored and a losers-only estimator drops the winner altogether. The resulting net price effects remain close: in non-pharmaceuticals, $p_{S_3} - p_{S_1}$ is 0.275/0.259/0.246 under losers-only/all-bidders/Turnbull inputs; in pharmaceuticals, the corresponding values are 0.347/0.308/0.357. The maximum Turnbull deviation from the baseline is 16 percent, and the within-auction component remains dominant: 74.0 percent in non-pharmaceuticals and 82.0 percent in pharmaceuticals.

The same conclusion survives the other cost-recovery diagnostics. The cross-modality check compares a Convite first-price recovery with the Pregão drop-out recovery in pre-policy cells that the set-aside does not target; after the auction-level heterogeneity correction, the recoveries line up in the key pharmaceutical non-SME pre-period cell. A Gaussian-copula relaxation of independent private values allows within-auction cost correlation up to $\rho_c = 0.3$; across that grid, the within-auction share moves by less than 5 percentage points and the total effect by less than 10 percent. Sample filters and windows also do not change the ordering. The baseline $c_\epsilon \leq 3$ filter preserves 99.6 percent of the data; tightening it lowers the total effect by 15/8 percent in non-pharmaceuticals/pharmaceuticals, while loosening it raises the effect by only 1/6 percent. Across 18-, 12-, and 6-month windows, the within-auction component remains the larger component in both classes, with share variation of only 4.3 and 6.6 percentage points.

6.3. Primitive stability and SME composition

The most important modeling threat is not whether entry responds; it does. The threat is that $S_3 - S_2$ combines two objects: more SME bidders and a changed post-policy SME cost distribution. The baseline specification treats the post-policy SME pool as the observed equilibrium participant pool. The strict-primitive-invariance benchmark imposes the opposite limiting case: $F_c^{\text{SME,Post}} = F_c^{\text{SME,Pre}}$, while keeping the observed post-policy SME entry counts.

That benchmark preserves the main mechanism. The simulated total price effect is still positive, at 0.29 in non-pharmaceuticals and 0.47 in pharmaceuticals, and the within-auction shares rise to 85 and 79 percent. Thus, even when the post-policy SME primitive is forced to be the pre-policy primitive, the price increase is still mainly the effect of excluding non-SMEs from the price-forming pool.

What changes is the policy ranking in the thin pharmaceutical market. In non-pharmaceuticals, the price-preference benchmark continues to dominate the full set-aside under both the main and strict-invariance readings. In pharmaceuticals, the ranking is conditional: the main specification favors the preference, while strict invariance reverses the ranking and lowers the implied SME welfare weight to 0.7. The turnover evidence explains why this is the right boundary of the paper rather than a cosmetic robustness issue. Post-policy pharmaceutical SME participation is heavily composed of new firms: 61.9 percent of post-period pharma SME firms are absent from the pre-period pool, and they account for 37.8 percent of post-period pharma SME bids. In non-pharmaceuticals, new firms account for 23.0 percent of post-period SME bids. The composition shift is real, especially in pharmaceuticals; the paper therefore treats the pharmaceutical welfare ranking as informative but conditional, and treats the non-pharmaceutical ranking as the robust policy result.

6.4. Coordination

Bid coordination is the threat that would most directly contaminate the cost-recovery exercise. If the post-policy restricted auctions induced surviving SMEs to coordinate more intensely, the recovered cost distributions could embed a conduct change rather than only the order-statistic effect of bidder exclusion. The data do show bidder clustering, so this concern cannot be waved away. Conley-style close-pair screens (Conley and Decarolis, 2016) reject the independent draw null in all four class-by-period cells, and persistent-pair diagnostics in the spirit of Bajari and Ye (2003) also flag excess repeated co-bidding.

The relevant question, however, is whether clustering increases after the restriction. It does not. The Conley realized close-pair share is essentially flat in non-pharmaceuticals (16.9 to 16.8 percent) and falls in pharmaceuticals (27.6 to 24.4 percent). The Bajari–Ye T_1 observed-to-null ratio falls from 2.63 to 1.83 in non-pharmaceuticals and from 1.29 to 1.11 in pharmaceuticals. This pattern is inconsistent with the strongest alternative story: a post-restriction coordination shock among the surviving SME pool. Residual baseline clustering remains a limitation, but it does not explain why the set-aside raises prices at the cutoff or why the within-auction component dominates the entry-composition offset.

6.5. Robustness of the policy comparison

The policy conclusion also survives the two mechanical concerns that could make the price-preference benchmark look artificially good. First, the welfare ranking is not driven by the marginal cost of public funds: over $\lambda \in [0.15, 0.45]$, the non-pharmaceutical ranking remains $V_3 \succ V_0$ under both the main and strict-invariance specifications, while the pharmaceutical ranking remains conditional on the treatment of the post-policy SME pool. Second, the 10 percent price preference is not relying on exact Vickrey equivalence. A Maskin–Riley first-price upper-bound adjustment (Maskin and Riley, 2000) is at most 5 percent of the reference price, while the simulated V_0–V_3 price gap is 22–26 percent. The ranking in thick standardized markets is therefore not a knife-edge artifact of auction format.

Taken together, the checks leave a narrow and defensible claim. The paper does not require the reduced-form DiD to estimate the mechanism, does not require a single cost-recovery estimator, and does not require post-policy SME entry to be ignored. What remains is the design principle: when the excluded bidders are the ones disciplining the order statistic, set-asides are costly because they replace competition with eligibility. SME entry offsets part of that cost, but in the observed markets it does not replace the competitive discipline that the policy removes.

7. Conclusion

SME set-asides are usually evaluated as a question of access: do protected firms receive more public contracts? This paper evaluates the instrument as a question of market design. A set-aside supports SMEs by changing eligibility, and changing eligibility changes price formation. It removes some bidders from the auction at the same time that it may attract protected firms into the auction. The welfare case for the policy depends on the balance of those two forces.

The evidence from São Paulo's procurement reform is clear on that balance. SME participation responds after the set-aside expands, but the response does not replace the competitive discipline supplied by excluded non-SMEs. The main price effect comes from the restricted auction's order statistic: once non-SMEs are removed, the price-forming bidder is drawn from a thinner and different pool. Entry and composition partly offset the shock, but they do not undo it. The mechanism is therefore not that SMEs simply bid less aggressively. It is that an exclusion rule changes who remains available to discipline the price.

That mechanism changes the relevant policy comparison. The benchmark is not SME support versus no SME support. It is exclusionary support versus support that preserves competition inside the auction. A full set-aside is a high-cost, high-redistribution instrument because it forces the award toward SMEs by removing rival bidders. A price preference is lower redistribution, but it keeps non-SMEs in the auction and allows them to remain price-forming. In thick standardized markets, that distinction is decisive: the paper's robust policy result is that preserving competition dominates removing it.

The conclusion is deliberately conditional. In thin or heterogeneous markets, especially where the post-policy protected pool changes substantially, the case for exclusion can be stronger and the ranking can become model-sensitive. That is not a failure of the design principle. It is the principle's boundary. A planner who chooses bidder exclusion should be explicit that the additional SME surplus, market-access objective, or dynamic capacity gain is worth the allocative and fiscal cost of taking rival bidders out of the auction.

The broader lesson is simple: eligibility is an expensive substitute for competition when the excluded firms are the ones disciplining the auction. Procurement policy should therefore start from instruments that support SMEs while keeping rival bidders in the room, and reserve full set-asides for markets where the planner is prepared to pay for the missing competition.

Declaration of competing interests

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The structural sample is derived from BEC administrative microdata accessed through a research agreement with the Secretaria da Fazenda e Planejamento do Estado de São Paulo. Public replication materials will be organized for the v8 version once the analysis set is frozen.

References

Athey, S., D. Coey, and J. Levin (2013). Set-asides and subsidies in auctions. *American Economic Journal: Microeconomics 5*(1), 1–27.

Athey, S., J. Levin, and E. Seira (2011). Comparing open and sealed bid auctions: Evidence from timber auctions. *Quarterly Journal of Economics 126*(1), 207–257.

Bajari, P. and L. Ye (2003). Deciding between competition and collusion. *Review of Economics and Statistics 85*(4), 971–989.

Ballard, C. L., J. B. Shoven, and J. Whalley (1985). General equilibrium computations of the marginal welfare costs of taxes in the united states. *American Economic Review 75*(1), 128–138.

Conley, T. G. and F. Decarolis (2016). Detecting bidders groups in collusive auctions. *American Economic Journal: Microeconomics 8*(2), 1–38.

Finkelstein, A. and N. Hendren (2020). Welfare analysis meets causal inference. *Journal of Economic Perspectives 34*(4), 146–167.

Guerre, E., I. Perrigne, and Q. Vuong (2000). Optimal nonparametric estimation of first-price auctions. *Econometrica 68*(3), 525–574.

Haile, P. A. and E. Tamer (2003). Inference with an incomplete model of English auctions. *Journal of Political Economy 111*(1), 1–51.

Hendren, N. and B. Sprung-Keyser (2020). A unified welfare analysis of government policies. *The Quarterly Journal of Economics 135*(3), 1209–1318.

Hyytinen, A., S. Lundberg, and O. Toivanen (2018). Design of public procurement auctions: Evidence from cleaning contracts. *The RAND Journal of Economics 49*(2), 398–426.

Krasnokutskaya, E. (2011). Identification and estimation of auction models with unobserved heterogeneity. *Review of Economic Studies 78*(1), 293–327.

Krasnokutskaya, E. and K. Seim (2011). Bid preference programs and participation in highway procurement auctions. *American Economic Review 101*(6), 2653–2686.

Marion, J. (2007). Are bid preferences benign? the effect of small business subsidies in highway procurement auctions. *Journal of Public Economics 91*(7–8), 1591–1624.

Maskin, E. and J. Riley (2000). Asymmetric auctions. *Review of Economic Studies 67*(3), 413–438.

Nakabayashi, J. (2013). Small business set-asides in procurement auctions: An empirical analysis. *Journal of Public Economics 100*, 28–44.

Saez, E. and S. Stantcheva (2016). Generalized social marginal welfare weights for optimal tax theory. *American Economic Review 106*(1), 24–45.